

AI CLINICAL ROUTER

MedTriage

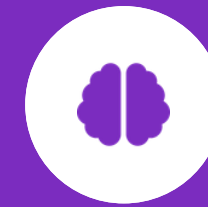
Symptom-to-department triage using NLP

Srushti Lambdade | 25030421021

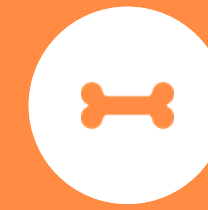
Individual project - NLP CA3: Build-Audit-Pitch
Symbiosis Artificial Intelligence Institute (SAII)
Faculty: Dr. Nabanita Choudhury | October 2026



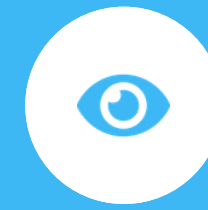
Cardiology



Neurology



Orthopedics



Ophthalmology



Pediatrics



Gastroenterology

Problem Statement

THE PROBLEM

Given a free-text clinical description, route the patient to the right one of 11 departments, and say how sure the model is.



Wrong first desk delays care



Input is unstructured text



11 departments to choose from

OBJECTIVES



Build

Working TF-IDF pipeline



Audit

Check subgroups and bias



Explain

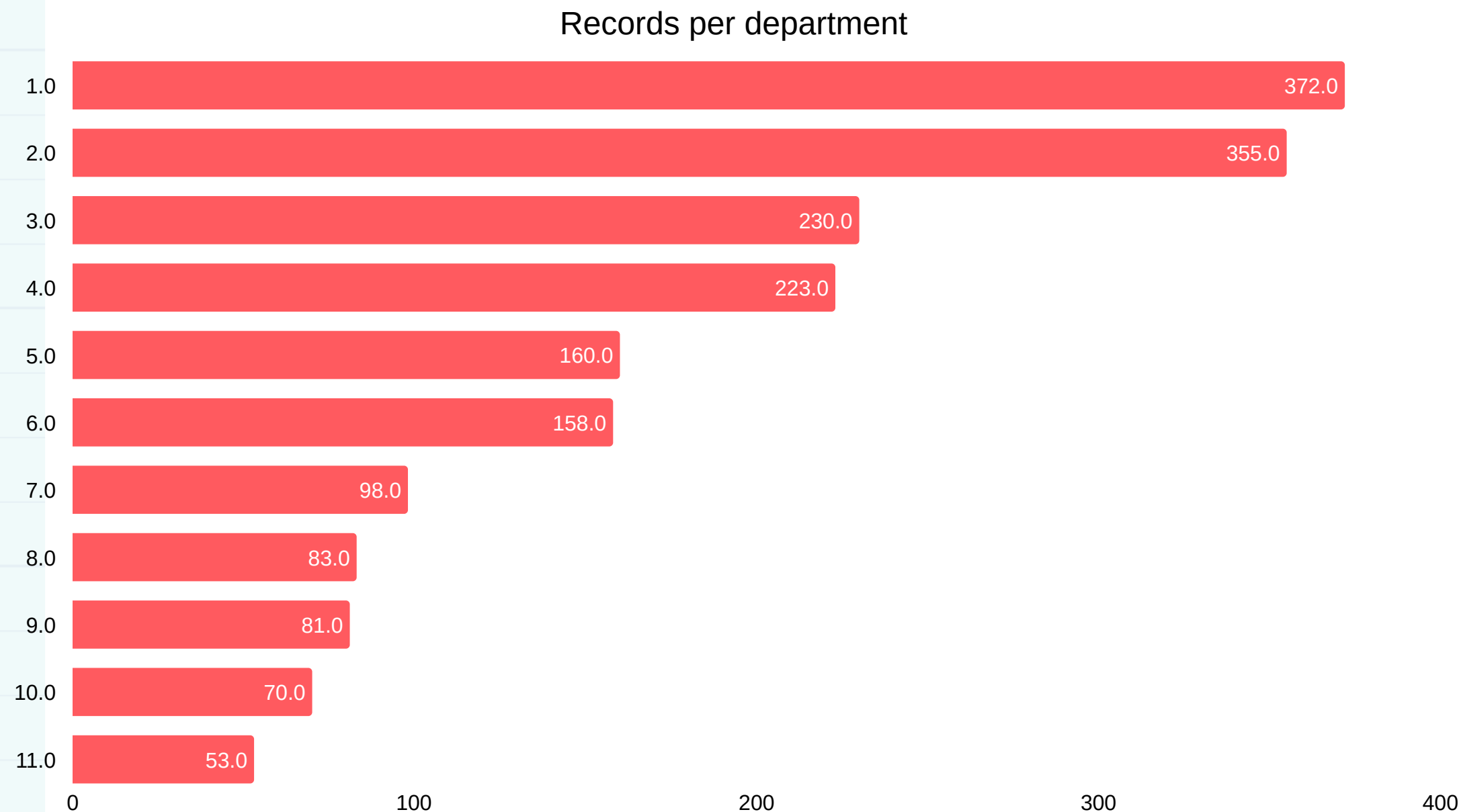
Top features per department



Trust

Calibrated confidence

Dataset: MTSamples



4,999

raw medical transcriptions

1,883

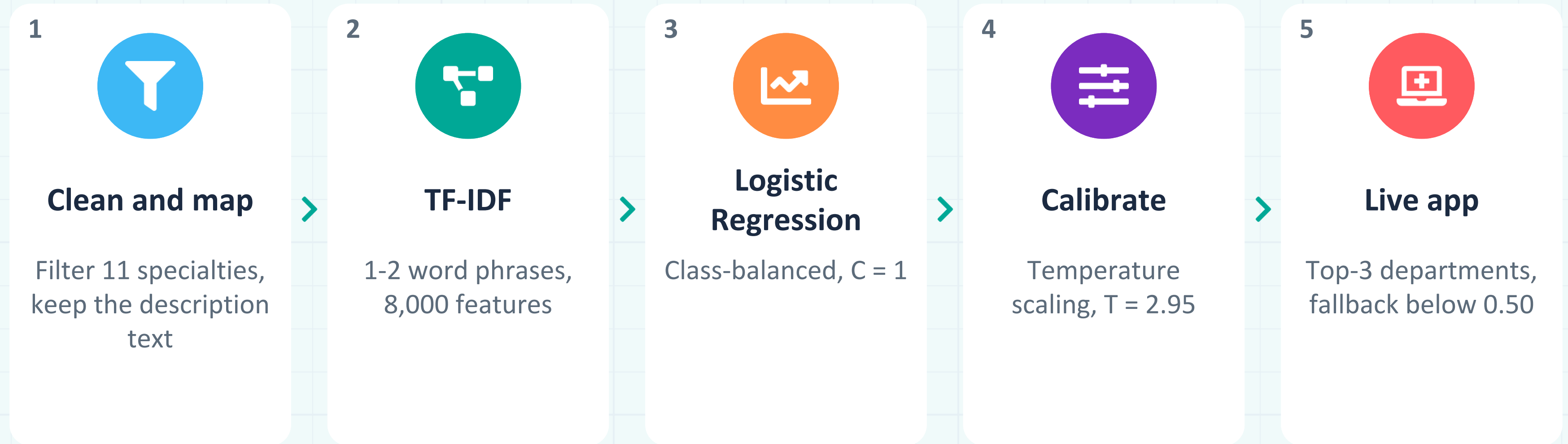
kept across 11 departments

1,506 / 377

train / test (80/20, stratified)

Public, de-identified, English | Cardiology has 7x more records than Psychiatry

Approach: the NLP pipeline



1.9 ms
per prediction

Interpretable
coefficients show why

Light
runs on a laptop

Results on unseen data

80%

accuracy (79.8%) vs 19.6% baseline

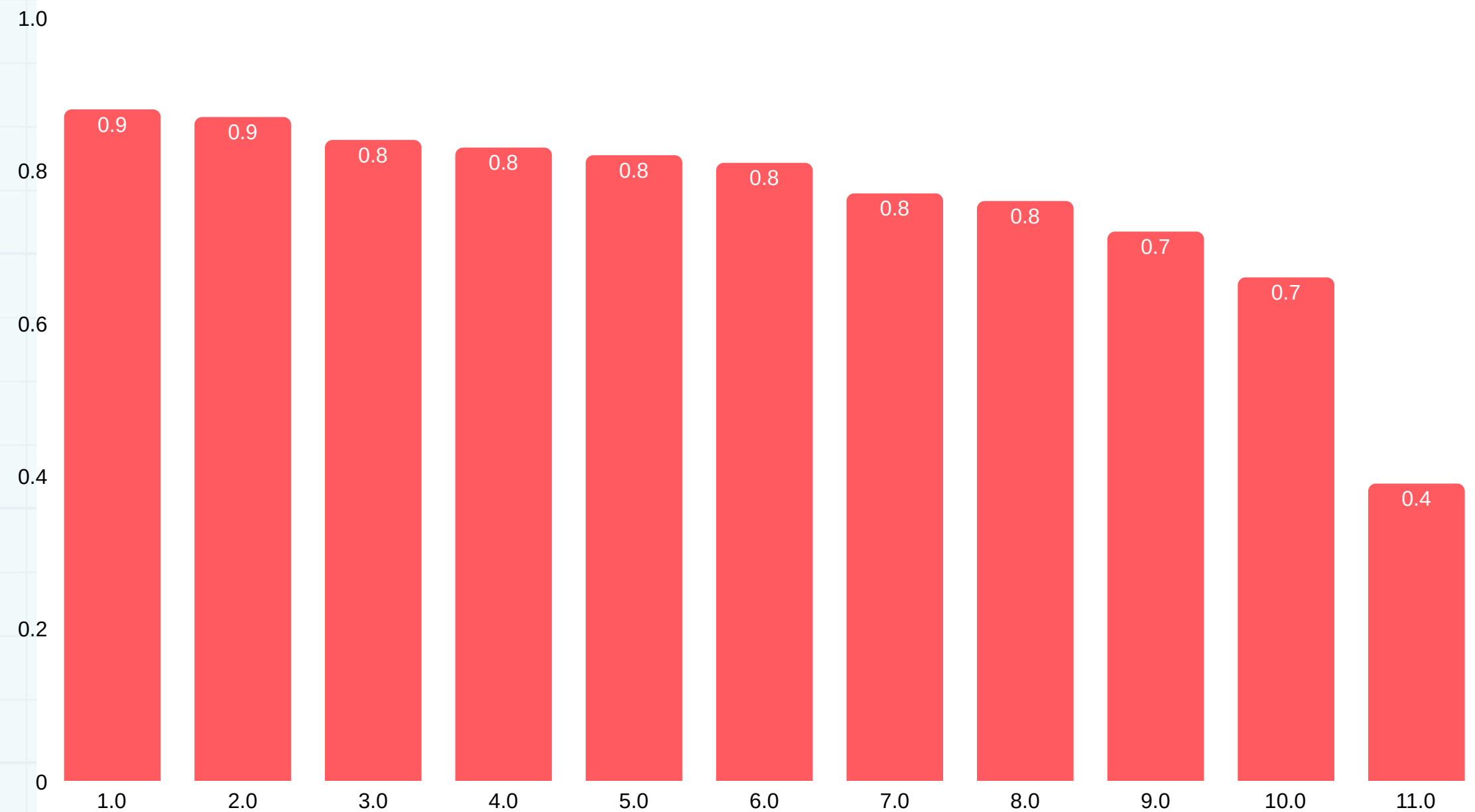
0.76

macro-F1 over 11 departments

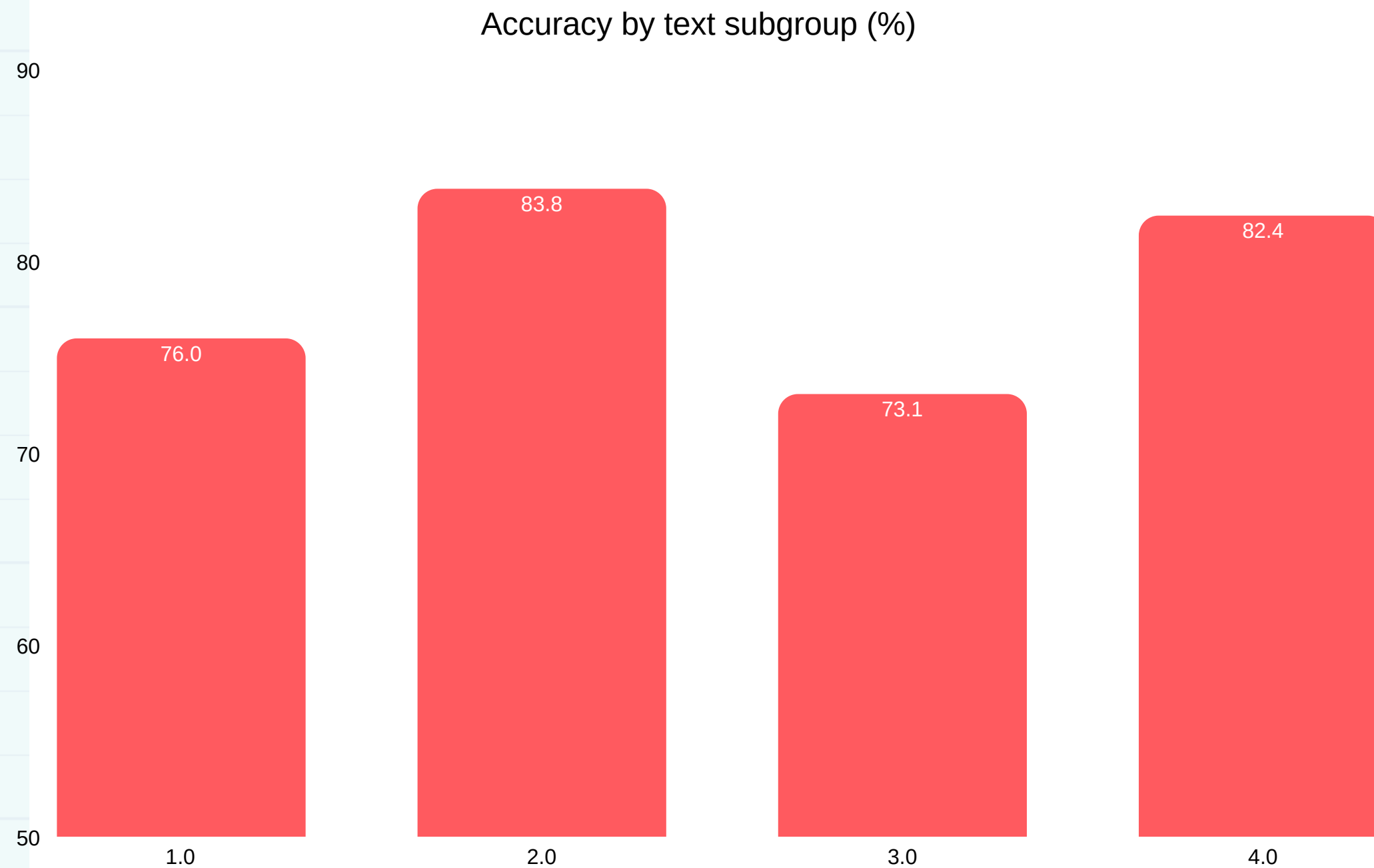
377

held-out test records

F1 score by department



Bias audit: where does it struggle?



KEY AUDIT FINDING

Short text loses 7.7 points; narrative text loses 9.3.

Narrative cases are full of generic words. Procedure titles carry specific cues like "cataract".

Weakest class: Pediatrics, F1 0.39

Explainability and honest limitation

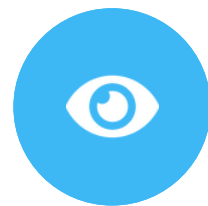


Cardiology

chest

coronary

pulmonary



Ophthalmology

eye

cataract

vitrectomy



Urology

bladder

prostate

inguinal



Pediatrics

child

old

month

generic



HONEST LIMITATION

**Word matching only,
no context.**

"4-month-old with tachycardia"
goes to Pediatrics (99%): age
words outweigh the cardiac
term.

Not for clinical use.

Highest-weight words from the Logistic Regression coefficients

Fixing the low confidence

BEFORE (raw model)

39%

chest-pain case, correct class: Cardiology



T = 2.95

AFTER (calibrated)

93%

same input, same prediction, honest score

ECE (calibration error)

0.426 -> 0.079

Mean confidence

0.37 -> 0.78

Accuracy

80% unchanged

Cause: sparse TF-IDF + C = 1 gives flat scores. **Fix:** scale scores, fit on training data only; below 0.50 goes to human triage.



Live Demo

MedTriage Gradio app

Type a symptom description and read the top-3 departments with calibrated confidence.

1 Clear case

"sharp chest pain radiating to the left arm with shortness of breath"

Cardiology 93%

2 Specific case

"blurry vision and redness in the left eye for three days"

Ophthalmology 100%

3 Vague case

"feeling tired and unwell for a few days"

Low confidence, human triage

Thank You

Questions are welcome

Srushti Lambdade | 25030421021

NLP CA3 - SAIL - October 2026



**80% accuracy and
0.76 macro-F1**



**Short and narrative text is
weakest**



**Calibrated confidence plus
human fallback**